

1. Test Execution Summary

Area	Tests	Result
API root / health	2	Passed
Input validation + demo generation	2	Passed
TXT/DOCX/PDF exporters	3	Passed
DOCX bold rendering + TXT Markdown stripping	2	Passed
Text sanitization / term parsing / filename safety	3	Passed
Safe HTML preview + inline Markdown	2	Passed
Total	14	14 passed

2. Test Cases

ID	Scenario	Expected	Status
TC-01	GET /	200 and LegalEase message	PASS
TC-02	GET /health	status=ok and model metadata	PASS
TC-03	Blank required field	422 validation error	PASS
TC-04	Demo generation	200 and document content returned	PASS
TC-05	TXT export	Starts with clean document text	PASS
TC-06	DOCX export	Valid DOCX with terms table	PASS
TC-07	PDF export	Valid %PDF header	PASS
TC-08	DOCX bold Markdown	Bold run is preserved	PASS
TC-09	TXT Markdown markers	Markers removed	PASS
TC-10	sanitize_text	Punctuation/control cleanup works	PASS
TC-11	terms_to_list	Semicolon parsing works	PASS
TC-12	safe_filename	Normalized file name generated	PASS
TC-13	HTML preview safety	Script tags escaped	PASS
TC-14	HTML Markdown	Bold/italic tags rendered	PASS

3. Test Method

- The API tests use FastAPI TestClient to exercise endpoints without starting an external server.
- Exporter tests inspect returned byte formats and, for DOCX, reopen the generated package with python-docx.
- Utility tests verify deterministic text-processing behavior.
- Demo generation is tested with provider access disabled by temporarily switching settings to demo_mode=True.

4. Defect and Quality Checklist

Check	Evidence / expectation
API validation	Pydantic rejects empty required data and unexpected request fields.
AI provider failure	Retry and fallback logic is present for temporary provider errors.

Check	Evidence / expectation
HTML injection	Preview escapes HTML before Markdown rendering.
Export correctness	TXT/DOCX/PDF bytes validated by automated tests.
Readable document	Final PDFs should be rendered and visually checked before submission.

5. Release Gate

Release gate for academic submission: automated tests pass, source repository contains the required phase folders, demo mode can be used for an offline presentation path, and the eight final PDFs have been visually inspected after rendering.